

HIMANSHU PAITHANE

New York, NY | P: 201-241-9114 | hpaithan@stevens.edu | [My Portfolio](#) | [LinkedIn](#)

WORK EXPERIENCE

PANASONIC NORTH AMERICA

May 2025 – Present

Data Engineer

- Developed and deployed a **Databricks (PySpark)** pipeline to transform and standardize **300,000+ customer** names using **NLP** + fuzzy matching, ranking candidate matches to improve downstream analytics reliability and reduce false matches.
- Built and deployed a scalable PySpark data validation framework to automate **ETL** reconciliation and migration QA, improving reporting trust and strengthening data-quality controls to support ongoing maintenance of downstream models and analytics.
- Automated **GA4 ETL** pipelines into **BigQuery** for product engagement and lifecycle analytics, then integrated curated datasets into **Power BI** to deliver dashboards that reduced reporting turnaround time and improved stakeholder decision-making.

JAR

Mar 2024 – Aug 2024

Data Scientist

- Developed and deployed an internal ML-based affluence scoring model (**LightGBM/XGBoost**) to classify new users at signup into spending-affinity tiers, validated on holdout data and operationalized deployment with **MLflow**, replacing a third-party solution and saving **~\$15k/month**.
- Developed a **Kafka-to-AWS S3** ingestion pipeline for transactional SMS events from **~20K DAUs**, orchestrated Airflow batch transforms and Redshift loads for analytics/modeling, and maintained operational datasets in MongoDB for downstream application use.
- Employed **NLP** techniques to process unstructured SMS data, converting it into structured formats for storage in **MongoDB**, enabling downstream analytics of user behavior and engagement.

SYMPHONY AI

May 2023 – Aug 2023

Data Science Intern

- Built and optimized a neural network for motor health prediction using sensor-derived **time-series** features, improving accuracy by **23%** through feature engineering and hyperparameter tuning, and delivering anomaly/risk scores to support maintenance decisions.
- Optimized production ML models (**SVM, XGBoost, Neural Networks**) through advanced feature engineering, statistical analysis, and hyperparameter tuning, boosting **F1-score to 90%** (confusion matrix/ROC-based evaluation) and enabling more reliable predictions at scale.
- Worked across **end-to-end ML** workflows including data preprocessing, feature engineering, model training, and validation across diverse model families (tree-based, classical ML, deep learning) to deliver reliable, production-oriented predictions.

SHARP

Jun 2021 – Sept 2021

Data Science Intern

- Built end-to-end machine learning frameworks using Python libraries (**PyTorch, Scikit-Learn, TensorFlow, Keras**) performed preprocessing, feature engineering, and hyperparameter tuning to deliver high-accuracy, production-ready models
- Designed, executed, and analysed large scale **A/B tests** to evaluate business strategies and product features, applying statistical methods to generate actionable insights that directly guided product decisions and user engagement improvements.
- Optimized big data pipelines using **Apache Spark & Hadoop** to process large datasets, boosting **efficiency by 4x** and enabling scalable real-time analytics

PROJECTS (Check out more [here](#))

VectorSearch RAG (MongoDB) - LangChain Chunking + Sentence-Transformers + PyMongo - [Link](#)

- Built an end-to-end **RAG** retrieval pipeline by chunking PDFs, generating local sentence-transformer embeddings, and storing vectors + metadata in MongoDB Atlas.
- Implemented **MongoDB Vector Search** indexing and \$vectorSearch queries to retrieve top semantically similar chunks with tuned candidate limits.
- Automated ingestion + retrieval in a reproducible Python workflow enabling retrieval-grounded (RAG) responses and secure, repeatable experimentation.

Multimodal Document Parsing Pipeline - OpenAI Responses API, Vision, PyMuPDF, Pydantic - [Link](#)

- Built an **LLM**-based document extraction pipeline using the **OpenAI Responses API** (vision) to convert PDF tables into structured records and export clean CSVs for downstream analytics.
- Iterated prompt + schema design (JSON schema + Pydantic validation) to improve extraction accuracy and consistency for classification tasks.
- Added repeatability and error-checking via multi-run consistency - hash comparisons to reduce variance and surface edge cases for targeted fixes.

FindMySquad - Full-Stack Web Development - [Link](#)

- Developed and deployed a full-stack web platform (**Node.js, Express.js, MongoDB, Handlebars, CSS, JavaScript**) for hosting/joining games, finding gym buddies, and organizing esports tournaments.
- Integrated real-time chat (**Websockets**), email reminders (**Nodemailer + Cron**), and interactive maps (**Leaflet.js**) to enhance collaboration and event coordination.
- Designed and implemented user engagement systems (profiles, ratings, karma leaderboard, achievements) to boost retention and fair play.

EDUCATION

STEVENS INSTITUTE OF TECHNOLOGY

Masters of Science in Computer Science (GPA: 3.9/4.0)